Replication Plan

Deformation Behavior of Annealed $Cu_{64}Zr_{36}$ Metallic Glass via Molecular Dynamics Simulations

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Auto-generated Replication Blueprint

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Contents

Paper Overview

This paper uses classical molecular dynamics (MD) simulations to study the deformation behavior of annealed $Cu_{64}Zr_{36}$ metallic glass. The workflow involves: (1) generating an amorphous Cu–Zr alloy via melt-quenching, (2) annealing the glass at different temperatures below T_g, (3) performing uniaxial tensile/compressive deformation, and (4) analyzing stress–strain curves, shear band formation, Voronoi polyhedral analysis of local structure, and free volume evolution. The study uses LAMMPS with the Finnis–Sinclair (EAM) interatomic potential for Cu–Zr.

Detailed Workflow

Phase 1: Sample Preparation

1.1 Build initial configuration.

- Create a random solid solution of 64% Cu and 36% Zr atoms in an FCC supercell.
- System size: ∼10,000–50,000 atoms (as specified in the paper).

1.2 Melt-quench protocol.

- Heat to 2000 K (well above melting point) and equilibrate in NPT.
- Quench to target temperature (e.g., 300 K) at specified cooling rate (e.g., 10^{10}–10^{12} K/s).
- Use Nosé–Hoover thermostat/barostat.

1.3 Annealing.

- Anneal at various sub-T_g temperatures for specified durations (e.g., 1–10 ns).
- Multiple annealing temperatures to study structural relaxation effects.

Phase 2: Deformation Simulations

2.1 Uniaxial tension/compression.

- Apply strain at constant engineering strain rate (e.g., 10^8–$10^9\,s^{-1}$).
- Use periodic boundary conditions in lateral directions.
- Record per-atom stress and strain tensors.

2.2 Record stress–strain curves.

- Compute virial stress tensor; extract σ_{zz} vs. ε_{zz}.
- Identify elastic modulus, yield stress, and ultimate strength.

Phase 3: Structural Analysis

3.1 Voronoi tessellation analysis.

- Compute Voronoi polyhedra indices $\langle n_3, n_4, n_5, n_6 \rangle$ for each atom.
- Track fractions of icosahedral ($\langle 0, 0, 12, 0 \rangle$) and other key polyhedra.

3.2 Free volume analysis.

- Compute atomic volume from Voronoi cells.

- Track free volume evolution during annealing and deformation.

3.3 Shear band visualization.

- Compute local atomic shear strain (η_i^{Mises}) using OVITO.
- Visualize shear band formation during deformation.

3.4 Radial distribution function $g(r)$ to characterize short-range order.

Phase 4: Results Reproduction

4.1 Reproduce stress–strain curves for different annealing conditions.

4.2 Reproduce Voronoi polyhedra fraction evolution plots.

4.3 Reproduce shear band snapshots.

4.4 Reproduce $g(r)$ plots and structural order parameter trends.

Datasets

Dataset / Source	**Type**	**Location / Notes**
Cu–Zr EAM potential	Interatomic potential	NIST Interatomic Potentials Repository `https://www.ctcms.nist.gov/potentials/` Specific potential: Mendelev et al. (2009) or as cited
Initial atomic configuration	Generated	Created by LAMMPS `create_atoms` with random alloy
Simulation parameters	From paper	Cooling rate, annealing T/time, strain rate, system size

Models, Tools, and Software

Simulation Software

Tool / Library	**Version**	**Purpose / URL**
LAMMPS	stable (2022+)	Molecular dynamics simulation engine `https://www.lammps.org/`
OVITO	≥ 3.7	Visualization, Voronoi analysis, atomic strain `https://www.ovito.org/`
OVITO Python API	≥ 3.7	Scripted analysis pipeline `pip install ovito`

Tool / Library	Version	Purpose / URL
Python (NumPy, SciPy)	≥ 3.8	Post-processing, stress–strain curve analysis
Matplotlib	≥ 3.4	Plotting stress–strain, $g(r)$, Voronoi statistics
Atomsk	latest	Optional: atomic structure manipulation `https://atomsk.univ-lille.fr/`

Interatomic Potential

- **Cu–Zr EAM/Finnis–Sinclair potential**: Mendelev et al. (2009) or the specific potential cited in the paper. Available from the NIST Interatomic Potentials Repository.

Hardware Requirements

- Multi-core workstation or small HPC cluster.
- GPU-accelerated LAMMPS (KOKKOS package) recommended for large systems.
- ≥ 16 GB RAM; $\sim$100 GB storage for trajectory data.